ET4382 Power Conversion Techniques in CMOS Technology – SMPC Project

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First Calculations

Duty Cycle: $\frac{V_{in}}{V_o} = Duty \ \Rightarrow \frac{1.2}{3.3} = D = 0.3636$

Now, for $I_L = 10mA, with\, V_o = 1.2V, R_L = \frac{V_o}{I_L} = 120\Omega$

Similarly, for $I_L = 10\mu A\ case, R_L = 1.2M\Omega$ (But this will become relevant later)

$F_{sw}, L\ and\ C\ Sizing$

$K = 1 - D = \frac{2LF_{sw}}{R_L}$ (Derivation in Appendix)

So, $LF_{sw} = 38.184$

And, Ripple Current is given by: $\Delta I_L = \frac{(V_{in} - V_{O})}{L * F_{sw}} * D$

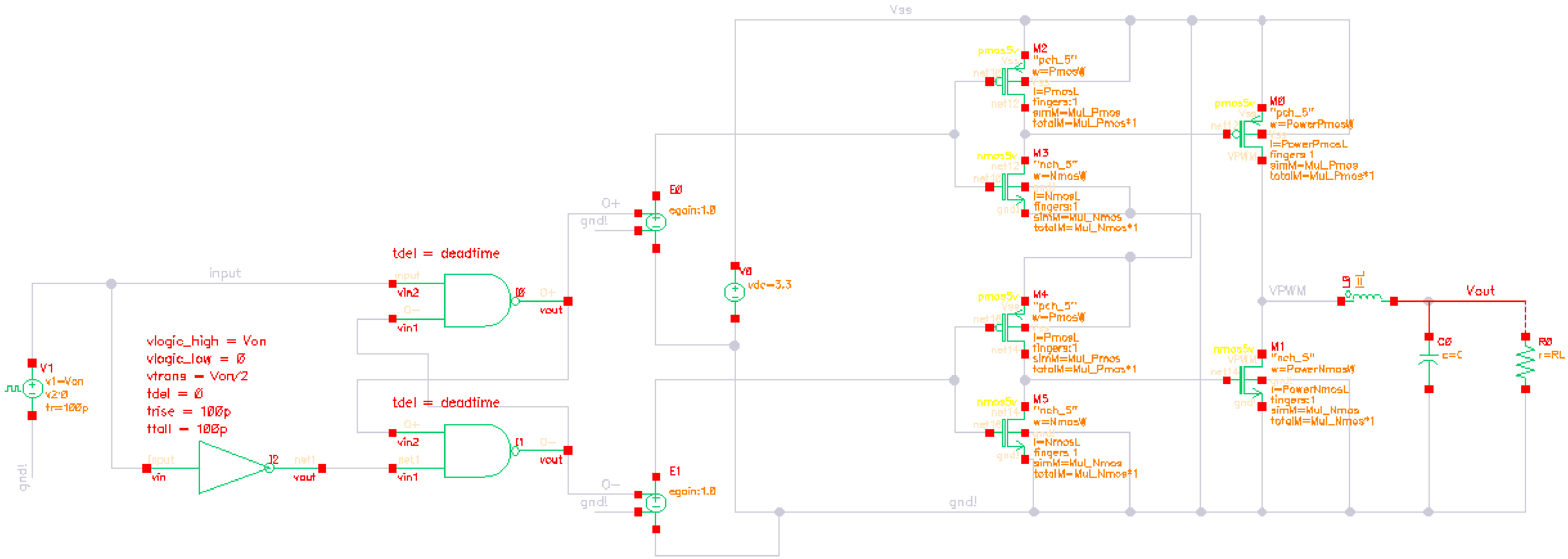
So, $\Delta I_L = 20mA$ (for any combination of $LF_{sw} = 38.184$)

Now, we look at the ripple voltage spec given (100mV):

$I = C\frac{\Delta V}{\Delta T} \Rightarrow C * F_{sw} = 0.2$

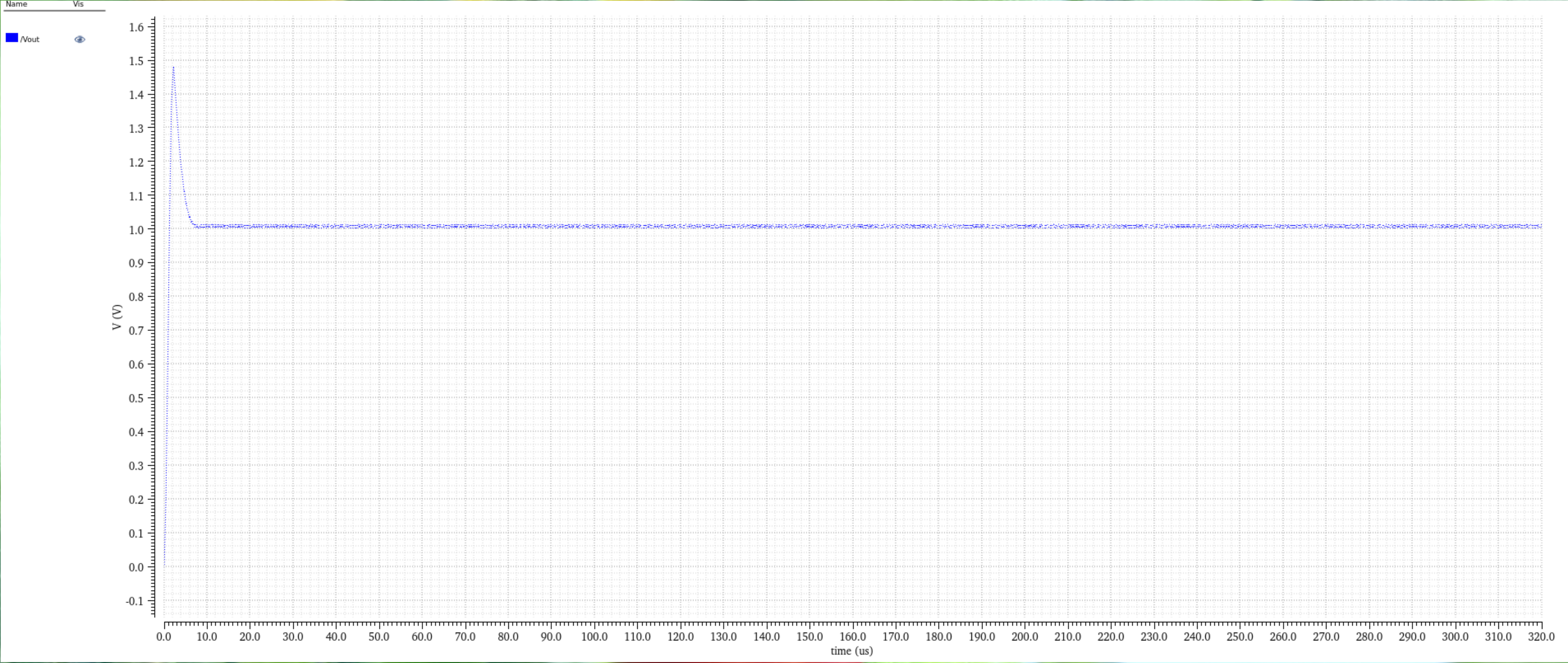
So we have $C * F_{sw} = 0.2$ & $LF_{sw} = 38.184$

Lets set $F_{sw} = 4MHz, so\ C = 50nF\ and\ L = 9.546\mu H$

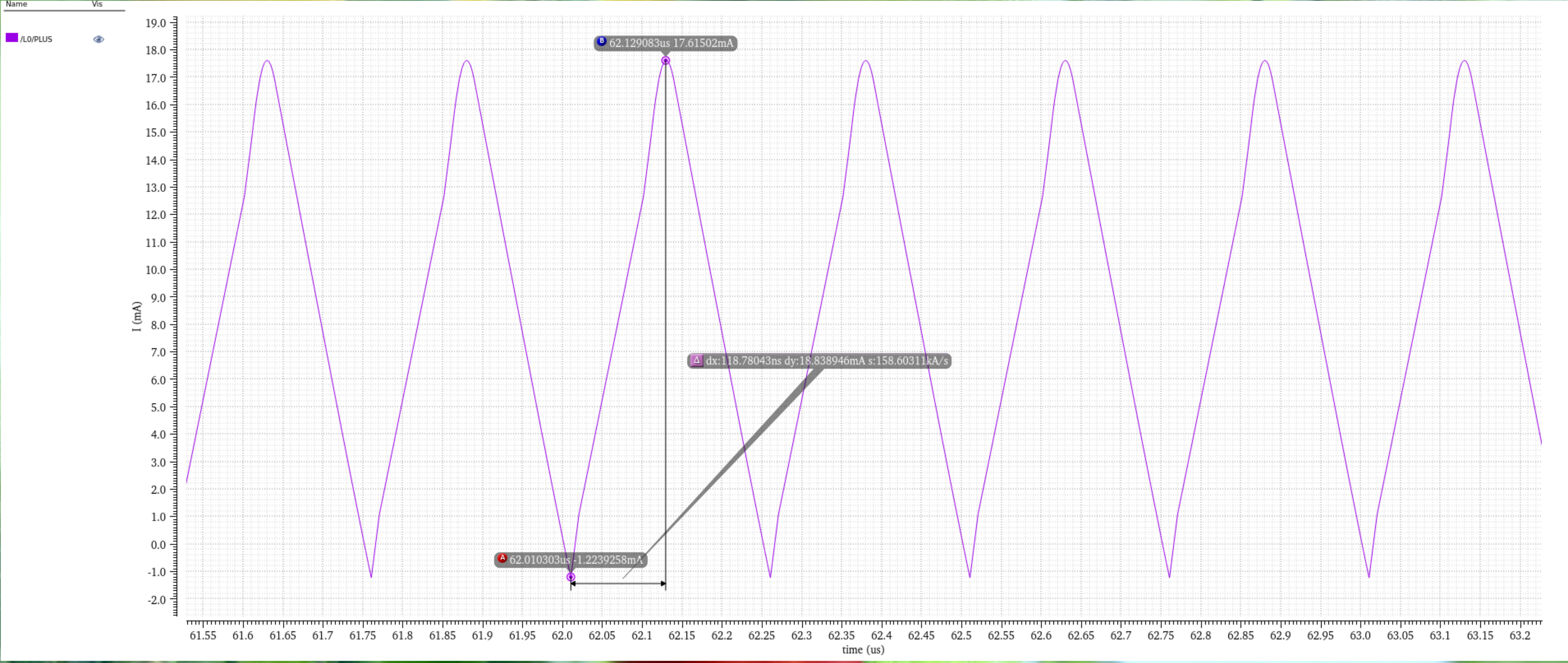


Vss
M2
"pch_5"
w=PmosW
l=PmosL
fingers:1
simM=Mul_Pmos
totalM=Mul_Pmos*1
M3
"nch_5"
w=NmosW
l=NmosL
fingers:1
simM=Mul_Nmos
totalM=Mul_Nmos*1
M0
"pch_5"
w=PowerPmosW
l=PowerPmosL
fingers 1
simM=Mul_Pmos
totalM=Mul_Pmos*1
E0
egain:1.0
V0
vdc=3.3
tdel = deadtime
input
vin2
vin1
vout
vlogic_high = Von
vlogic_low = 0
vtrans = Von/2
tdel = 0
trise = 100p
tfall = 100p
V1
v1=Von
v2:0
tr=100p
vin
vout
net1
E1
egain:1.0
M4
"pch_5"
w=PmosW
l=PmosL
fingers:1
simM=Mul_Pmos
totalM=Mul_Pmos*1
M5
"nch_5"
w=NmosW
l=NmosL
fingers 1
simM=Mul_Nmos
totalM=Mul_Nmos*1
M1
"nch_5"
w=PowerNmosW
l=PowerNmosL
fingers:1
simM=Mul_Nmos
totalM=Mul_Nmos*1
VPWM
Vout
C0
c=C
R0
r=RL
gnd!

Schematic


Name
Vis
/Vout
V (V)
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
-0.1
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
110.0
120.0
130.0
140.0
150.0
160.0
170.0
180.0
190.0
200.0
210.0
220.0
230.0
240.0
250.0
260.0
270.0
280.0
290.0
300.0
310.0
320.0
time (us)

Vout Plot



Ripple Current Plot

Power FET and Gate Driver Sizing

Size the power FETs to trade off switching loss and conduction loss for high peak efficiency

For that, we do a calculation for $P_{cond} = P_{avg} + P_{ripple}$

For finalized W/L values for Nmos and Pmos: 400u/600n and 400u/500n

Calculated Ron values for Nmos and Pmos are 203m and 72m respectively

So, Calculated Efficiency is = 99.3%

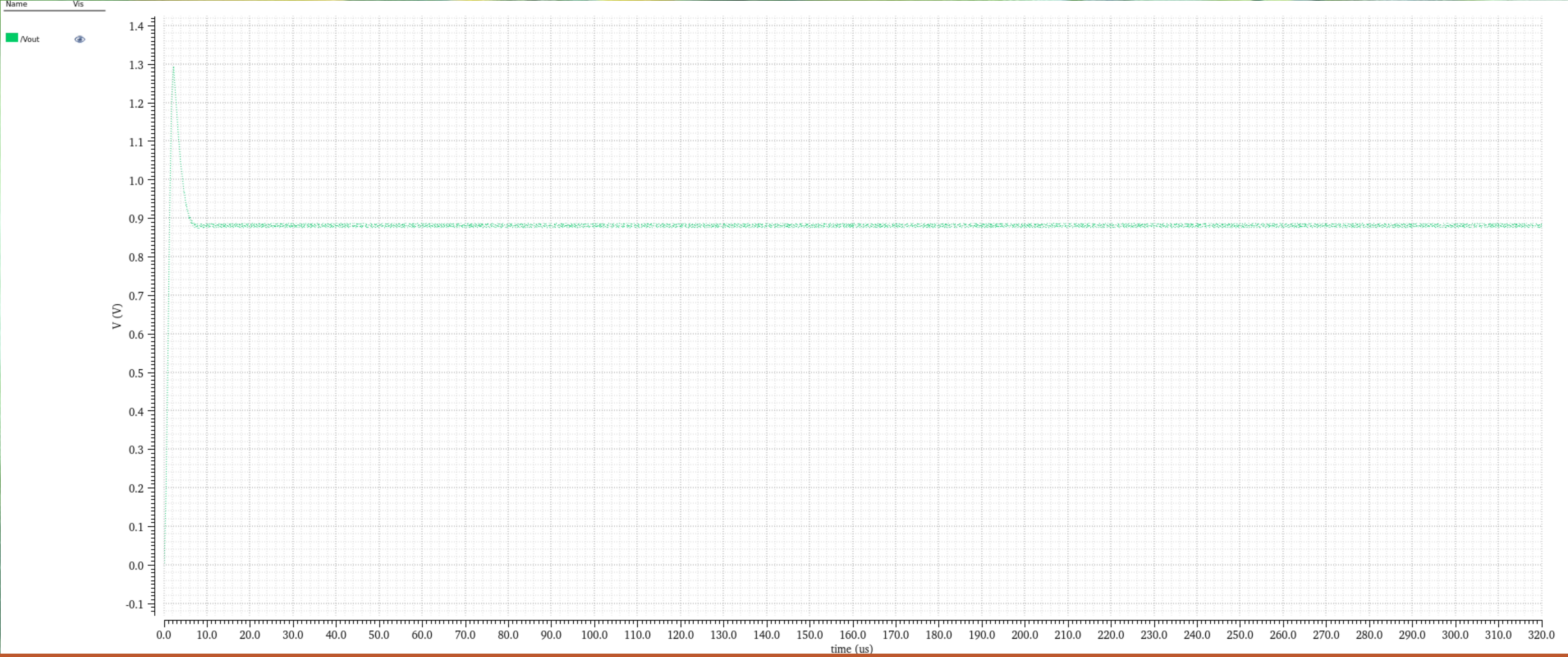
Gate Drivers were sized Iteratively to balance transition loss and ringing due to large di/dt.

Impact of parasitic inductances

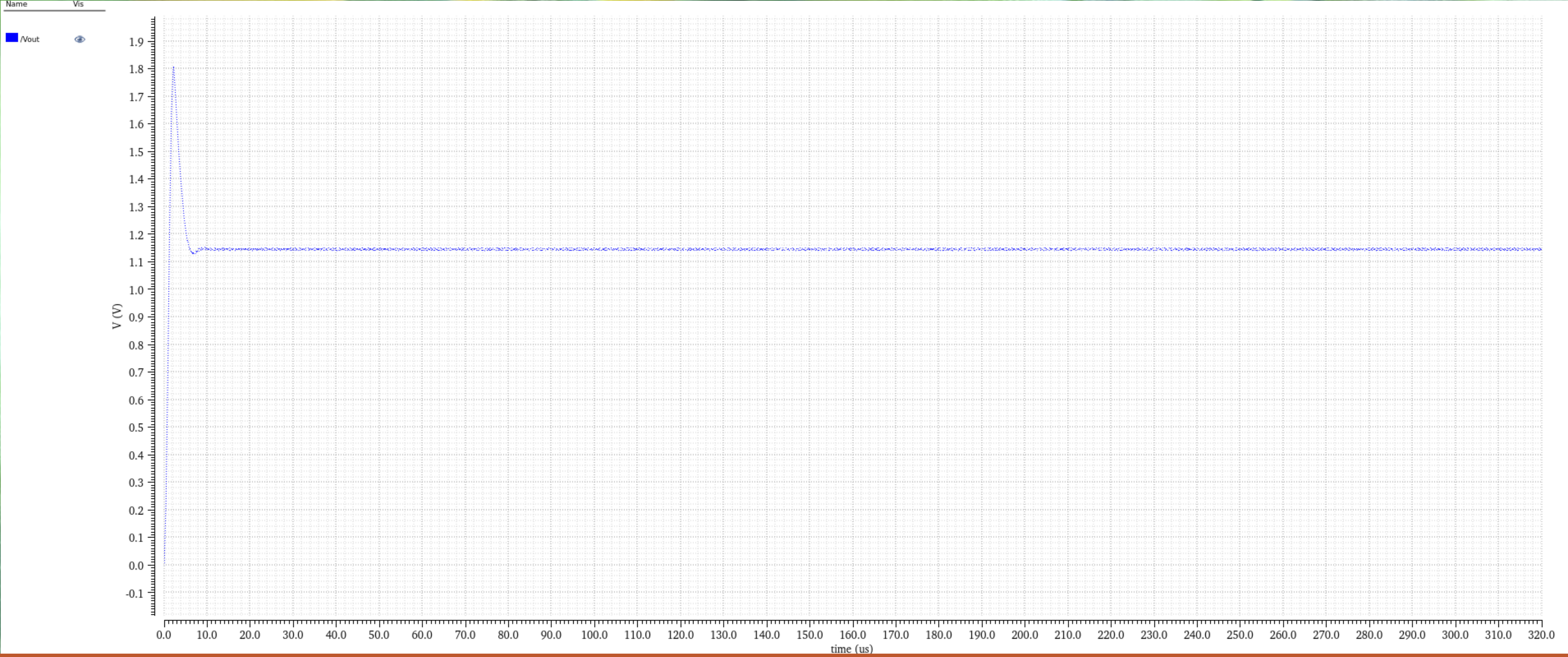
There will be parasitic effect from any external connection. So we modelled them as an Inductor in series with a resistance:

The external battery connection

And the R_L



Impact of parasitic inductances



Final Vout Plot without control loop

Thank you!

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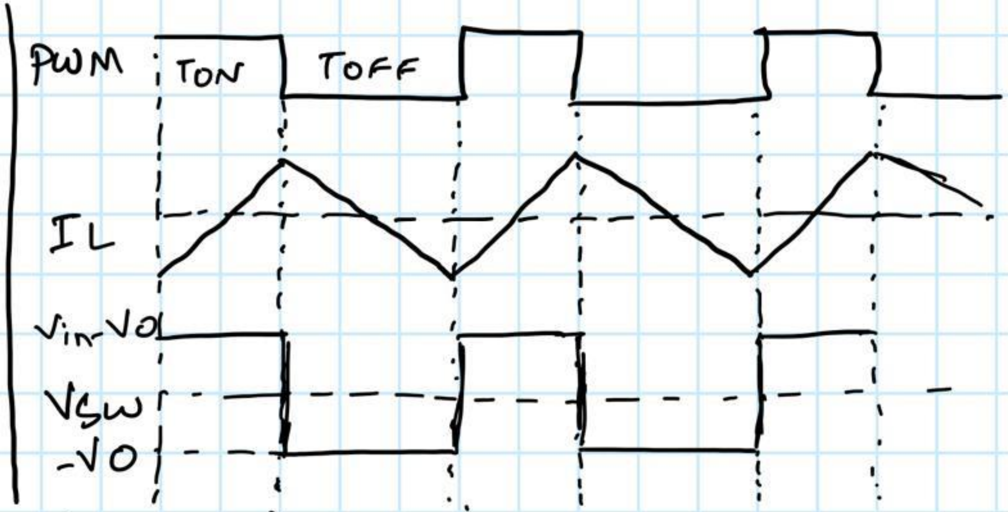
Appendix - 1

CCM-DCM Boundary condition:

$$I_{Load} = \frac{\Delta I_L}{2} = \frac{1}{2}\frac{V_O}{L}\frac{(1-D)}{F_{SW}}$$

$$I_{Load} = \frac{V_O}{R_L}$$



So,

$$(1-D) = \frac{2LF_{SW}}{R_L} = K \text{ (critical point)}$$

Appendix - 2

$R_{on,pmos} = 72\,m\Omega$

$R_{on,nmos} = 203\,m\Omega$

$I_{avg} = 8.572\,mA$; $Duty = 0.3636$; $I_{ripple} = 18.84\,mA$

$P_{cond} = P_{avg} + P_{ripple}$

$$P_{avg} = I^2_{avg} R_{on,pmos} Duty + I^2_{avg} R_{on,nmos} (1-Duty)$$

$$= I_L^2{}_{avg} \times R_{eff} = (8.572m)^2 \times (0.155)$$

$$= 1.14 \times 10^{-5}\,W$$

$$P_{ripple} = I^2_{ripple} R_{on,pmos} Duty + I^2_{ripple} R_{on,nmos} (1-Duty)$$

$$= (18.84\,m)^2 (72\,m)(0.3636) + (18.84m)^2 (203m)(0.6364)$$

$$= (18.84m)^2 (0.155) = 5.5 \times 10^{-5}\,W$$

$$P_{cond} = P_{avg} + P_{ripple} = 6.64 \times 10^{-5}\,W$$

$$\text{total power} = I(L_{avg}) \times V_L$$

$$= 8.572\,m \times 1.2 = 0.0103\,W$$

$$\frac{P_{cond}}{P_{total}} = 6.45 \times 10^{-3} \Rightarrow \boxed{\eta = 99.3\%}$$